

HDOC DAY-38

Title: String Analysis: Character Category Counting, Specific Frequency Analysis, and Case Toggling

1. Problem Statement

Write a menu-driven C program to perform the following operations using separate functions:

1. Count spaces, digits, and special characters in a string.
2. Count frequency of a given character in a string.
3. Toggle the case of each character in a string.
4. Exit.

2. Theoretical Background & Logic

1. Character Categorization:

By inspecting the ASCII values of each character in the array:

- **Space:** `str[i] == ' '`
- **Digits:** `'0' <= str[i] <= '9'`
- **Alphabets:** `'a' <= str[i] <= 'z' or 'A' <= str[i] <= 'Z'`

- **Special Characters:** Any character that does not fit into space, digit, or alphabet categories.

2. Frequency Count:

A target character is provided by the user. A single-pass loop traverses the string comparing `str[i] == target` and increments a counter variable whenever a match is found.

3. Case Toggling Logic:

In ASCII encoding, corresponding lowercase and uppercase letters differ by exactly 32:

- Uppercase to Lowercase: $ch + 32$ (e.g., 'A' (65) + 32 = 97 ('a')).
- Lowercase to Uppercase: $ch - 32$ (e.g., 'a' (97) - 32 = 65 ('A')).

3. Test Case Table

Test Case	Selected Option	Input String	Expected Output	Actual Output	Status
1	Count Categories	"A1 @b"	Spaces: 1, Digits: 1, Special:	Spaces: 1, Digits: 1, Special:	Pass

Tes t Ca se	Selecte d Option	Input String	Expecte d Output	Actual Output	Stat us
			1, Alphabe tic: 2	1, Alphabe tic: 2	
2	Freque ncy Analysi s	"banan a", Char: 'a'	Frequen cy of 'a': 3	Frequen cy of 'a': 3	Pass
3	Toggle Case	"C Progra m #1"	"c pROGR AM #1"	"c pROGR AM #1"	Pass
4	Exit	Option 5	Termina te program executio n	Termina ted program executio n	Pass

4. Source Code Listing

```
#include <stdio.h>

void read_string(char str[]) {
```

```

printf("Enter a string: ");
scanf(" %[^\\n]", str);
}

void count_spaces_digits_special(const char str[]) {
    int spaces = 0, digits = 0, special = 0, alphabets = 0;
    int i = 0;

    while (str[i] != '\\0') {
        if (str[i] == ' ') {
            spaces++;
        } else if (str[i] >= '0' && str[i] <= '9') {
            digits++;
        } else if ((str[i] >= 'a' && str[i] <= 'z') || (str[i] >= 'A' && str[i] <= 'Z')) {
            alphabets++;
        } else {
            special++;
        }
        i++;
    }

    printf("\\n--- Character Category Counts ---\\n");
    printf("Spaces      : %d\\n", spaces);
    printf("Digits       : %d\\n", digits);
    printf("Special Characters : %d\\n", special);
    printf("Alphabets      : %d\\n", alphabets);
}

void count_character_frequency(const char str[]) {
    char target;
    int count = 0, i = 0;

    printf("Enter character to search frequency for: ");
    scanf(" %c", &target);

    while (str[i] != '\\0') {
        if (str[i] == target) {
            count++;
        }
        i++;
    }
}

```

```

    }

    printf("\nFrequency of '%c' in string : %d\n", target, count);
}

void toggle_case(const char input[], char output[]) {
    int i = 0;

    while (input[i] != '\0') {
        if (input[i] >= 'a' && input[i] <= 'z') {
            output[i] = input[i] - 32;
        } else if (input[i] >= 'A' && input[i] <= 'Z') {
            output[i] = input[i] + 32;
        } else {
            output[i] = input[i];
        }
        i++;
    }
    output[i] = '\0';
}

int main(void) {
    char str[200];
    char result[200];
    int choice;

    read_string(str);

    do {
        printf("\n===== STRING OPERATIONS MENU\n\n");
        printf("1. Count spaces, digits, and special characters\n");
        printf("2. Count frequency of a given character\n");
        printf("3. Toggle case of each character\n");
        printf("4. Enter a new string\n");
        printf("5. Exit\n");
        printf("Enter your choice (1-5): ");
        scanf("%d", &choice);

        switch (choice) {

```

```
    case 1:
        count_spaces_digits_special(str);
        break;

    case 2:
        count_character_frequency(str);
        break;

    case 3:
        toggle_case(str, result);
        printf("\nOriginal String : %s\n", str);
        printf("Toggled String : %s\n", result);
        break;

    case 4:
        read_string(str);
        break;

    case 5:
        printf("Exiting program...\n");
        break;

    default:
        printf("Error: Invalid choice! Please select between 1 and 5.\n");
        break;
}
} while (choice != 5);

return 0;
}
```

4. Execution Screenshots

```
kali@kali:~/HDOC_Day38
Session Actions Edit View Help
(kali@kali)~$ mkdir HDOC_Day38
cd HDOC_Day38
(kali@kali)~/HDOC_Day38$ nano day38.c
(kali@kali)~/HDOC_Day38$ gcc day38.c -o day38
(kali@kali)~/HDOC_Day38$ ./day38
Enter a string: Hello World!@#

===== STRING OPERATIONS MENU =====
1. Count spaces, digits, and special characters
2. Count frequency of a given character
3. Toggle case of each character
4. Enter a new string
5. Exit
Enter your choice (1-5): 1

--- Character Category Counts ---
Spaces       : 1
Digits       : 0
Special Characters : 3
Alphabets    : 10

===== STRING OPERATIONS MENU =====
1. Count spaces, digits, and special characters
2. Count frequency of a given character
3. Toggle case of each character
4. Enter a new string
5. Exit
Enter your choice (1-5): 2
Enter character to search frequency for: l

Frequency of 'l' in string : 3

===== STRING OPERATIONS MENU =====
1. Count spaces, digits, and special characters
2. Count frequency of a given character
3. Toggle case of each character
4. Enter a new string
5. Exit
Enter your choice (1-5): 3

Original String : Hello World!@#
Toggled String  : hELLO wORLD!@#

===== STRING OPERATIONS MENU =====
1. Count spaces, digits, and special characters
2. Count frequency of a given character
3. Toggle case of each character
4. Enter a new string
5. Exit
Enter your choice (1-5):
```